3C: OJ purchases and score-based decisions

Problem statement and worked resolution

Companion to `03_C_oj_score_thresholds.ipynb`

How to use this explanation. This resolution explains the prediction setting, score-threshold search, and why a policy can lower cost while making more classification mistakes. Section numbers follow the notebook's numbered sections where present. Numerical values are rounded and refer to the recorded run, not universal performance claims.

Problem statement and scope

The textbook OJ data contain 1,070 purchases: 653 CH and 417 MM. Encode MM as class 1 and CH as 0. Predict from PriceCH, PriceMM, DiscCH, DiscMM, SpecialCH, and SpecialMM. The first four describe prices and discounts, and the last two promotion indicators. The target Purchase never appears among the inputs. This is a specified restricted-feature benchmark, not a claim that the six variables contain every useful predictor.

The SVM learns a decision score $f(x)$. A policy predicts MM when $f(x) \geq t$. The costs are assumed: predicting MM for a CH purchase costs 1; predicting CH for an MM purchase costs 3. They could represent asymmetric consequences in a teaching scenario, but are not estimated from the records. We first select an SVM, then select its threshold using a different data partition.

1 Load data and reserve partitions

The notebook uses the public textbook repository URL with a 20-second network timeout. This avoids requiring local course files in Colab. Shape, label-count, and missing-value checks verify the expected data. The OJ data documentation provides variable context.

Stratification with seed 23 produces 642 training, 214 validation, and 214 test rows. Training contains 392 CH and 250 MM records. Price columns have substantially different numerical ranges from discount and binary promotion columns; each pipeline standardizes the six columns using fitting rows only. The same stores occur in every partition and their week ranges overlap. Consequently this random-split result concerns similarly sampled records. It does not show how a model would perform in an unseen store or in future weeks, and row-index separation alone cannot establish independence of repeated purchases.

2 Select model settings without test information

For logistic, linear SVM, and RBF SVM, tune C in $\{0.01, 0.1, 1, 10\}$; RBF also tunes gamma in $\{0.01, 0.1, 1, 10\}$. Each pipeline includes its own fitting scaler. Five training folds have slightly different sizes because 642 is not divisible by five: held-out folds have 128 or 129 rows, with the remaining 514 or 513 used to fit.

Mean balanced accuracy gives equal weight to CH recall and MM recall. It does not minimize the later cost directly. That separation is deliberate: fitting a useful score and choosing a cost-sensitive cutoff answer different questions.

Family	Selected C	Gamma	CV balanced accuracy
Logistic	0.1	—	0.622797
Linear SVM	1	—	0.582513
RBF SVM	1	0.1	0.630041

The selected SVM is the RBF model. Logistic regression remains a tuned comparison, not a compulsory loser. All three models are refitted on the full training partition and left unchanged thereafter. The chosen model's class order is checked as $[0, 1]$, so positive SVM scores favor MM.

3 Inspect scores and search every decision pattern

There are only 64 distinct validation scores, giving 65 possible threshold decision patterns. Repeated or equivalent input records can receive identical scores. A threshold cannot split a tied-score group into different labels without introducing another decision rule.

Let sorted distinct scores be $s_1 < \cdots < s_k$. Under the rule $f \geq t$, use cutoffs

$$-\infty, s_2, s_3, \ldots, s_k, +\infty.$$

The first predicts MM for every record, the last CH for every record, and each intermediate cutoff excludes one more lowest-score group. This covers all patterns without arbitrary fine grid spacing. The exact magnitude of a representative cutoff is less important than the group of records it classifies positive.

For every pattern, count false positives FP and false negatives FN and minimize $FP + 3FN$. Division by 214 converts this to mean cost but does not change its minimum. Among equal-cost cutoffs, use the largest, which favors fewer positive predictions. Including extreme policies prevents overlooking an optimum that always predicts one class.

The selected cutoff is approximately -1.001416, with validation cost $117/214 \approx 0.546729$. Always predicting MM has cost 130; always predicting CH has cost $3 \times 84 = 252$ on validation. The plot displays finite cutoffs only, but the extremes remain in the optimization and printed table. The

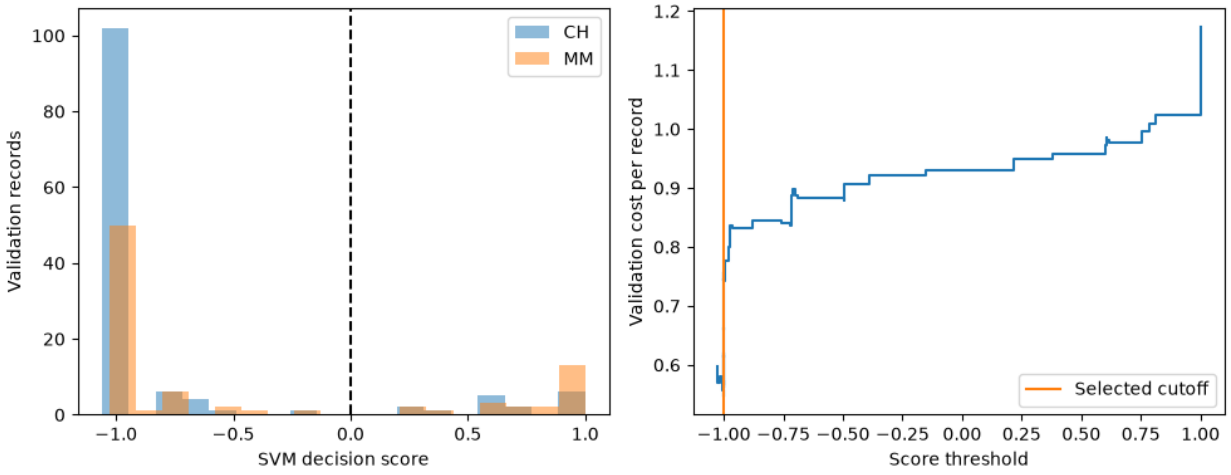


Figure 1: Validation score distributions and threshold costs. The zero cutoff is a default score convention; the selected cutoff reflects this sample and the specified costs.

cost curve is piecewise constant between score values. Connecting or stepping between plotted values is a visualization convention, not evidence that costs vary smoothly with every tiny threshold movement.

4 Why the probability threshold 1/4 does not apply to scores

For a true conditional MM probability p, the expected cost of predicting MM is $1 \times (1 - p)$, and the cost of predicting CH is $3 \times p$. Prefer MM when $1 - p \leq 3p$, hence $p \geq 1/4$. This derivation concerns a probability scale. A raw SVM score can be negative and has no such scale. The selected score near -1 is therefore not inconsistent with the probability argument.

Applying $\sigma(f) = 1/(1 + e^{-f})$ without learning a calibration mapping would merely transform numbers; it would not establish that they equal purchase risks. A strictly increasing transform preserves score ordering and the possible threshold policies, but changes the numerical cutoff. For example, transforming both scores and their selected cutoff monotonically can preserve every decision without improving calibration at all.

After threshold selection, retain the fitted model and scaler. Refitting them on training plus validation could change scores and invalidate the cutoff, even if the new model had similar overall accuracy. The focused experiment does not perform calibration or report probability losses for these raw scores.

5 Final test results and the cost–accuracy tradeoff

At the default zero-score policies:

Model	Accuracy	Balanced accuracy	ROC AUC
Training-majority	0.6122	0.5000	0.5000
Logistic	0.6402	0.5935	0.6509
Linear SVM	0.6495	0.5769	0.6480
RBF SVM	0.6776	0.6197	0.6731

ROC AUC evaluates ranking from real scores, not the selected cutoff, and does not require calibrated probabilities. It is distinct from both balanced accuracy and operational cost. For the selected RBF SVM:

Policy	TN	FP	FN	TP	Accuracy	Cost/record
Zero score	115	16	53	30	0.6776	0.8178
Validation cutoff	18	113	1	82	0.4673	0.5421

The changed cutoff prevents 52 misses while adding 97 false positives. The cost change is

$$97 - 3 \times 52 = -59,$$

so mean cost falls by $59/214 \approx 0.2757$ despite the increase in the ordinary error count. The selected policy has precision $82/195 \approx 0.4205$ and recall $82/83 \approx 0.9880$. This is a high-recall, many-false-alarm policy, not a universally better classifier.

For scale, always predicting MM on this test would cost $131/214 \approx 0.6121$; always predicting CH would cost $249/214 \approx 1.1636$. These transparent reference calculations show what an asymmetric-cost improvement means beyond a comparison with the default zero cutoff. They do not trigger reselection after test evaluation. The validation-selected policy remains fixed regardless of whether any reference happens to look better in another test sample.

Documentation and practical advice

The SVM documentation distinguishes decision scores from probability estimation. Check `classes_` when translating score signs, and keep the complete preprocessing pipeline when saving a policy. A threshold value without its fitted score function is not a reproducible classifier.

- Name the positive event before calculating FP, FN, precision, recall, or cost; reversing labels changes the interpretation.
- Search tied scores as groups and include the two constant policies. An arbitrary numeric grid may omit relevant decisions.
- A cost minimum fitted on validation has selection noise. Evaluate it once on the reserved test.
- Never interpret a price coefficient, support vector, or selected cutoff as evidence of a causal price intervention.
- For a future-store objective, change the evaluation design first; better random-split accuracy cannot answer that question on its own.